

IMANSHA DILSHAN

AI Engineer | Full-Stack Developer | AI Researcher

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PROFESSIONAL SUMMARY

Final-year Software Engineering undergraduate and AI Engineer with hands-on experience designing and deploying production grade Generative AI systems, including LLM-powered RAG pipelines, multi-agent frameworks, and multimodal AI services. Industry experience at WSO2 contributing to enterprise Generative AI initiatives, complemented by full-stack development across React, Spring Boot, FastAPI, and cloud platforms (Azure, GCP, Choreo). Passionate about building scalable, evaluation-driven AI products that solve real-world problems.

TECHNICAL SKILLS

Generative AI & ML	: LLMs, Prompt Engineering, RAG, Multi-Agent Systems (LangGraph, CrewAI), pgvector, Embedding Pipelines, Multimodal AI, HITL, AI Evaluation, Computer Vision (YOLOv8), LoRA Fine-Tuning
AI Frameworks	: FastAPI, LangChain, LangGraph, CrewAI, Streamlit, Gemini AI, GPT-4o, OpenAI API, Llama 3.2, Qwen2-VL-7B
ML & Deep Learning	: TensorFlow, Keras, Scikit-learn, LSTM, Autoencoders, Dense ANN
Backend	: Python, Java, Spring Boot, Node.js, Express.js, Ballerina, PHP
Frontend	: Next.js, React, TypeScript, JavaScript, Redux, Tailwind CSS, HTML5, CSS
Databases	: PostgreSQL, MySQL, MongoDB, pgvector, Redis
Cloud & DevOps	: AWS, Azure, GCP, WSO2 Choreo, Docker, Git, CI/CD, Vercel
Tools	: Figma, Adobe photoshop

EDUCATION

B.Sc. (Hons.) in Software Engineering | University of Kelaniya

Jul 2023 - Present

CGPA: 3.92 / 4.00 · Kelaniya, Sri Lanka

· Specializing in Net-Centric Applications, Data Science & Engineering Applications, Health Informatics.

PROFESSIONAL EXPERIENCE

Intern Software Engineer - WSO2

Sep 2025 - Feb 2026

Solution Architect Team

Colombo, Sri Lanka

- Deployed Geronimo V2 with Gemini AI Grounded Search for real-time lead enrichment using production FastAPI services on WSO2 Choreo.
- Engineered Geronimo Phase 2 RAG system with document summarization, keyword extraction, pgvector search, and LLM-based response validation.
- Contributed shared drive support to the open-source Ballerina Google Drive Connector; released in ballerinax/googleapis.drive 3.4.1.
- Co-developed WSO2 AI Course content and led a session on securing agentic AI applications using Asgardeo IAM.

RESEARCH PUBLICATIONS

Agentic Generative AI Framework for Industrial Predictive Maintenance

Mar 2026 - Jun 2026

IEEE IES Generative AI Challenge 2026 | **Accepted · IRAI 2026, Melbourne, Australia**

LangGraph · LangChain · FastAPI · GPT-4o · pgvector · YOLOv8 · TensorFlow · LSTM · Python

- Six-node LangGraph multi-agent pipeline with a Physics-Aware Hybrid Confidence Layer achieving 90.15% precision at 3.35% FPR; end-to-end fault diagnosis in 6.8 s, 300x faster than the manual baseline across four industrial machine types.
- Multimodal RAG engine combining image-captioning embeddings with text retrieval, plus HITL continuous learning pipeline improving task accuracy without model retraining.

Autonomous Multi-Agent AI Platform for Real-Time Flood Disaster Response

Apr 2026 - Present

iPURSE 2026, Peradeniya University International Research Symposium | **Under Revision · Abstract**

CrewAI · Llama 3.2 · Qwen2-VL-7B · LoRA · GeoPandas · OSMnx · NetworkX · AMD MI300X

- Fine-tuned Qwen2-VL-7B via LoRA on AMD MI300X for satellite flood zone detection; 4 CrewAI agents with Llama 3.2 backbone orchestrate flood polygon generation, OSMnx road-block detection, and NetworkX hazard-avoidant evacuation routing.
- Produces complete situational awareness reports with optimal evacuation routes in under 60 seconds.

Inclusive Disaster Early Warning Systems for Cognitive Impairments

Mar 2026 - Present

SENG 43216 - Final Year Research, University of Kelaniya | **In Progress**

Supervised by Dr. Anuradha Madugalla (Deakin University) & Dr. Nalin Warnajith (University of Kelaniya)

- Human-centered requirements engineering for inclusive disaster early warning systems targeting users with dementia, autism, MCI, and ADHD; 10 evidence-based WCAG-mapped guidelines via systematic literature review.
- Participatory co-design workshops, MoSCoW requirements specification, and empirical usability validation (N=15–20) with Figma prototype and functional mobile MVP.

FEATURED PROJECTS

Industrial Copilot, Agentic AI Predictive Maintenance Platform

Mar 2026 - Present

FastAPI · Next.js · LangGraph · GPT-4o · TensorFlow · LSTM · pgvector · Docker

- Enterprise-grade 6-node LangGraph multi-agent pipeline for safety-critical diagnostics with a 4-stage AI validation engine (physics constraints + GPT-4o); Dense/LSTM autoencoders for per-machine anomaly detection.

[Live](#)

GeoRescue, Multi-Agent GIS Flood Disaster Response

Apr 2026 - Present

CrewAI · Qwen2-VL-7B · LoRA · GeoPandas · NetworkX · Llama 3.2 · AMD MI300X

- AMD Developer Hackathon: fine-tuned Qwen2-VL-7B via LoRA for satellite flood detection; 4-agent CrewAI orchestration produces situational reports in under 60 seconds.

[GitHub](#)

SalonSync, Microservices Salon Booking Platform(Backend)

Jun 2025 - Sep 2025

Spring Boot · React · MySQL · RabbitMQ · Keycloak · Stripe · Docker

- Microservices platform with Keycloak/JWT auth, Stripe payments, RabbitMQ async messaging, and full Docker containerization.

[GitHub](#)

ExamBuddy, Online Examination Platform

Jan 2026 - Mar 2026

FastAPI · Next.js · PostgreSQL · Redis · Celery · PayHere · Neon · Upstash (Proprietary)

- Full-stack educational platform with timed exams, auto-grading, admin dashboard, and PayHere payments on a serverless Neon PostgreSQL + Upstash Redis stack.

[Live](#)

Customer Churn Prediction, Deep Learning

Dec 2025

Python · TensorFlow · Keras · Scikit-learn · Streamlit

- ANN-based bank churn predictor with real-time Streamlit inference and color-coded risk indicators.

[GitHub](#)

[Live](#)

CERTIFICATES

Introduction to Generative AI (Google) · Introduction to Cloud 101 (AWS Educate) · React Basics (Meta) · Programming with JavaScript (Meta) · HTML and CSS in Depth (Meta) · Version Control (Meta)

LEADERSHIP & ACTIVITIES

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|---|---------------------|
| • President - Software Engineering Students' Association, University of Kelaniya | Jun 2025 - Jun 2026 |
| • Head of Design - IEEE WIE Student Branch Affinity Group, University of Kelaniya | Aug 2024 - Aug 2025 |
| • PR Lead, IT Unit - Leo Club of University of Kelaniya | Aug 2024 - May 2025 |

LANGUAGES

English (Professional Proficiency) | Sinhala (Native)

REFERENCES

Minura Kariyawasam

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Prof. (Mrs.) I. U. Hewapathirana

Head / Senior Lecturer, University of Kelaniya

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